

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 December 2024 - January 2025

22MCA102 – ADVANCED DATABASE SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) How to separate the user applications from the physical database? Explain with a neat diagram.	10	L*2	1	1
	b) Draw the ER diagram for the student examination system with entity types such as Student, Subject, Exam and Result. Define relationship, cardinality ratios and participation constraints and represent them in the ER diagram.	10	L3	1	1
2.	a) Explain intension and extension with respect to ER Modelling for the following examination system. Draw the schema diagram for the student examination system with entities such as Student, Subject, Exam, Result by representing necessary constraints.	10	L3	1	1
	b) What are the implications of using Database approach? Explain by comparing with the drawbacks of traditional file processing system.	10	L2	1	1
Unit – II					
3.	a) Consider the Following Relations Student(Id, Name, Age, Email) Subject(Sid, sname, credit) Exam(Eid, Date, Duration, Subject_id) Result(Rid, Marks_obtained, Student_id, Exam_id) Write the queries to the following i. Find all exams scheduled for a specific subject (e.g., "Mathematics"). ii. Retrieve a list of students who scored above 75 marks in any exam, including their names, exam dates, and the corresponding subject names. iii. List the students who have scored between 60 and 80 marks in any exam, along with their marks and the exam date. iv. List the exams that have a duration of more than 2 hours.	10	L3	2	1
	b) Explain the following aggregate functions with syntax and example. i. TO_CHAR() ii. TO_NUMBER() iii. TO_DATE()	10	L2	2	1
4.	a) Consider the Following Relations, Student(Student_Id, Name, Emailid, Semester) Industry(Industry_ID, Industry_name, Location) Project(ProjectId, Project_title, Industry_ID, Student_ID, Startdate, Enddate) Evaluation(Evaluation_ID, Project_Id, Internal_Marks, Final_Marks, Total_Marks)				

Write the queries to perform the following:

- i. Find the total number of students working in each industry.
 - ii. List all students who have completed their projects but haven't been evaluated yet
 - iii. Find students who have scored more than 85 marks (internal + final) in total and the projects they worked on.
 - iv. List all projects done in the industry located in 'Bangalore' and the corresponding students.
- b) With syntax and example, explain the different types of Outer Joins.

10	L3	2	1
10	L2	2	1

Unit – III

5. a) Consider the tables from Q.No 4. a) and answer the following
- i. Create a trigger that automatically sets the Total_Marks to 0 if both Internal_Marks and Final_Marks are less than or equal to 50 upon inserting or updating a record in the Evaluation table.
 - ii. Write a stored procedure that takes a Student_ID as input and returns the average of the Total_Marks for that student across all projects.
- b) Explain the mechanism in Oracle to implement implicit concurrency control.

10	L3	3	1
10	L2	3	1

6. a) Write a cursor to retrieve all projects along with the corresponding internal and final marks of students. The cursor should then calculate and display whether the student passed (Total Marks $\geq$ 50) or failed (Total Marks $<$ 50) for the table Project in Q.No 4.a)
- b) What is a trigger? Explain its types.

10	L3	3	1
10	L2	3	1

Unit – IV

7. a) Describe the different types of anomalies that occur in unnormalized tables.
- b) Given the following functional dependencies, find the candidate key(s):
Table: R(A, B, C, D)
Functional Dependencies:

A $\rightarrow$ B
B $\rightarrow$ C
A, D $\rightarrow$ C

8	L2	4	1
12	L3	4	1

8. a) For the following table, identify any potential transitive dependencies and remove them to achieve 3NF:

Order_ID	Product_Name	Customer_ID	Customer_Name
O001	Laptop	C001	Alice
O002	Phone	C002	Bob
O003	Tablet	C001	Alice

5	L1	4	1
---	----	---	---

- b) Write an algorithm to check if a decomposition is lossless. Apply the same to the following set of relations, decomposition and functional dependencies:

R={ssn, ename, pno, pname, ploc, hours}
R1={ssn, ename} R2={pno, pname, ploc} R3={ssn, pno, hours}
D={ R1, R2, R3 }
F = { ssn $\square$ ename, pno $\square$ {pname, ploc}, {ssn, pno} $\square$ hours }

15	L3	4	1
----	----	---	---

Unit – V

9. a) Consider the tables from Q.No 4. a) and write the mongoDB commands for the following.
- Update the email address of a specific student (e.g., Student_Id = 'S001').
 - Sort projects by start date in ascending order and limit the results to the top 5.
 - Find all evaluations with total marks greater than or equal to 70 and sort them by total marks in descending order.
 - Update the internal and final marks for a specific project (e.g., Project_Id = 3001) and set the Total_Marks to the sum of these marks.
 - Find the projects that belong to a specific industry (e.g., Industry_ID = 'I001') and sort them by project title in alphabetical order. Limit the result to 10.
- b) Write a note on
- ObjectID
 - Features of NoSQL database.
10. a) Consider the tables from Q.No 4. a) and write the mongoDB commands for the following.
- Update a Student's Internal Marks for a Specific Project
 - Delete All Students Who Have Not Been Assigned a Project
 - Find all evaluations where a student scored more than 60 in internal marks.
 - List all students who are working on projects for a specific industry (e.g., Industry_ID = 102).
 - Retrieve projects sorted by Project_Title in ascending order and limit the result to 4.
- b) Explain the different types of NoSQL databases.

10 L3 5 1

10 L1 5 1

10 L3 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
